

```
#include <conio.h>
```

```
#include <graphics.h>
```

```
#include <iostream>
```

```
#include <math.h>
```

```
#include <stdio.h>
```

```
int gd = DETECT, gm;
```

```
int n, x[100], y[100], i;
```

```
float sfx, sfy;
```

```
void draw();
```

```
void scale();
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    cout << "Enter no. of sides in polygon: ";
```

```
    cin >> n;
```

```
    cout << "Enter coordinates x, y for each vertex: ";
```

```
    for (i = 0; i < n; i++) {
```

```
        cin >> x[i] >> y[i];
```

```
    }
```

```
    cout << "Enter scale factors: sfx and sfy : ";
```

```
    cin >> sfx >> sfy;
```

```
    initgraph(&gd, &gm, (char*)"");
```

```
    cleardevice();
```

```
    setcolor(RED);
```

```
    draw();
```

```
scale();

setcolor(YELLOW);

draw();

getch();

closegraph();

return 0;

}

void draw()

{

    for (i = 0; i < n; i++) {

        line(x[i], y[i], x[(i + 1) % n], y[(i + 1) % n]);

    }

}

void scale()

{

    for (i = 0; i < n; i++) {

        x[i] = x[0] + (int)((float)(x[i] - x[0]) * sfx);

        y[i] = y[0] + (int)((float)(y[i] - y[0]) * sfy);

    }

}
```

```
#include<stdio.h>

#include<graphics.h>

#include<conio.h>

#include<iostream>

#include<math.h>

int gd= DETECT, gm;

int n,x[100],y[100],i;

float sfx, sfy;

void draw();

void scale();

using namespace std;

int main(){

cout<<"Enter no. of sides in polygon: ";

cin>>n;

cout<<"Enter coordinates x, y for each vertex: ";

for(i=0;i<n;i++){

cin>>x[i]>>y[i];}

cout<<"Enter scale factors: sfx and sfy : ";

cin>>sfx>>sfy;

initgraph(&gd, &gm, (char*)"");

cleardevice();

setcolor(RED);

draw();

scale();

setcolor(YELLOW);

draw();

getch();
```

```
closegraph();

return 0;

}

void draw(){

for(i=0; i<n; i++){

line(x[i],y[i],x[(i+1)%n],y[(i+1)%n]);}

}

void scale(){

for(i=0; i<n; i++){

x[i]=x[0]+(int)((float)(x[i]-x[0])*sfx);

y[i]=y[0]+(int)((float)(y[i]-y[0])*sfy);

}

}
```

```

#include<stdio.h>

#include<graphics.h>

#include<math.h>

#include<iostream>

using namespace std;

int main()

{

int gd=0,gm,x1,y1,x2,y2,x3,y3;

double s,c, angle;

initgraph(&gd, &gm, (char*)"");

setcolor(RED);

cout<<"Enter coordinates of triangle: ";

cin>>x1>>y1;

cin>>x2>>y2;

cin>>x3>>y3;

setbkcolor(WHITE);

cleardevice();

line(x1,y1,x2,y2);

line(x2,y2, x3,y3);

line(x3, y3, x1, y1);

cout<<"Enter rotation angle: ";

cin>>angle;

c = cos(angle *M_PI/180);

s = sin(angle *M_PI/180);

x1 = floor(x1 * c + y1 * s);

y1 = floor(-x1 * s + y1 * c);

x2 = floor(x2 * c + y2 * s);

```

```
y2 = floor(-x2 * s + y2 * c);  
x3 = floor(x3 * c + y3 * s);  
y3 = floor(-x3 * s + y3 * c);  
setcolor(GREEN);  
line(x1, y1, x2, y2);  
line(x2, y2, x3, y3);  
line(x3, y3, x1, y1);  
getch();  
closegraph();  
return 0;  
}
```

```

#include <conio.h>

#include <graphics.h>

#include <iostream>

#include <stdio.h>

int gd = DETECT, gm;

int n, xs[100], ys[100], i, ty, tx;

void draw();

void translate();

using namespace std;

int main()
{
    cout << "Enter no. of sides in polygon: ";

    cin >> n;

    cout << "Enter coordinates x, y for each vertex: ";

    for (i = 0; i < n; i++) {

        cin >> xs[i] >> ys[i];

    }

    cout << "Enter distances for translation (in x and y "

        "directions): ";

    cin >> tx >> ty;

    initgraph(&gd, &gm, (char*)"");

    cleardevice();

    setcolor(RED);

    draw();

    translate();

    setcolor(YELLOW);

    draw();

```

```
    getch();

    closegraph();

    return 0;
}

void draw()
{
    for (i = 0; i < n; i++) {
        line(xs[i], ys[i], xs[(i + 1) % n],
            ys[(i + 1) % n]);
    }
}

void translate()
{
    for (i = 0; i < n; i++) {
        xs[i] += tx;
        ys[i] += ty;
    }
}
```

